

Solution to Practice Problems 25–27

Practice Problem 25: C(1) and G(1) Continuity

Definitions:

- **C(1) continuity:** Both position and first derivatives are continuous
- **G(1) continuity:** Position is continuous and tangent directions are equal

a. At $t = 2\pi$

$$(x(t), y(t)) = \begin{cases} (t, \sin t) & \text{for } t \leq 2\pi \\ (t, 1 - \cos t) & \text{for } t > 2\pi \end{cases}$$

Position continuity:

Left: $(2\pi, \sin(2\pi)) = (2\pi, 0)$

Right: $(2\pi, 1 - \cos(2\pi)) = (2\pi, 1 - 1) = (2\pi, 0)$

✓ Position continuous

First derivatives:

Left: $(1, \cos t) \rightarrow (1, \cos(2\pi)) = (1, 1)$

Right: $(1, \sin t) \rightarrow (1, \sin(2\pi)) = (1, 0)$

✗ Derivatives not equal: $(1, 1) \neq (1, 0)$

Answer: C(0) continuous only (position only)

b. At $t = \pi/4$

$$(x(t), y(t)) = \begin{cases} (t, \sin t) & \text{if } t \leq \frac{\pi}{4} \\ (t, 1 - \cos t) & \text{if } t > \frac{\pi}{4} \end{cases}$$

Position continuity:

Left: $(\pi/4, \sin(\pi/4)) = (\pi/4, \sqrt{2}/2)$

Right: $(\pi/4, 1 - \cos(\pi/4)) = (\pi/4, 1 - \sqrt{2}/2)$

✗ Position not continuous: $(\pi/4, \sqrt{2}/2) \neq (\pi/4, 1 - \sqrt{2}/2)$

Answer: Not C(0) continuous (discontinuous at junction)

c. At $t = 1$

$$(x(t), y(t)) = \begin{cases} (6t, t^3) & \text{if } t \leq 1 \\ (t^4 + 5, t^2) & \text{if } t > 1 \end{cases}$$

Position continuity:

Left: $(6, 1)$

Right: $(1 + 5, 1) = (6, 1)$

✓ Position continuous

First derivatives:

Left: $(6, 3t^2) \rightarrow (6, 3)$

Right: $(4t^3, 2t) \rightarrow (4, 2)$

✗ Derivatives not equal: $(6, 3) \neq (4, 2)$

Tangent directions:

Left tangent: $(6, 3) \rightarrow$ direction ratio 2:1

Right tangent: $(4, 2) \rightarrow$ direction ratio 2:1

✓ Same direction (proportional vectors)

Answer: G(1) continuous

d. At $t = 1$

$$(x(t), y(t)) = \begin{cases} (t, t^2) & \text{for } t \leq 1 \\ (t, t^2 + (t - 1)^3) & \text{for } t > 1 \end{cases}$$

Position continuity:

Left: $(1, 1)$

Right: $(1, 1 + 0) = (1, 1)$

✓ Position continuous

First derivatives:

Left: $(1, 2t) \rightarrow (1, 2)$

Right: $(1, 2t + 3(t - 1)^2) \rightarrow (1, 2 + 0) = (1, 2)$

✓ Derivatives equal: $(1, 2) = (1, 2)$

Answer: C(1) continuous

Function	C(0)	G(1)	C(1)
25(a) at $t = 2\pi$	Yes	No	No
25(b) at $t = \pi/4$	No	No	No
25(c) at $t = 1$	Yes	Yes	No
25(d) at $t = 1$	Yes	Yes	Yes

Practice Problem 26: C(2) and G(2) Continuity

Definitions:

- **C(2) continuity:** Position, first and second derivatives continuous
- **G(2) continuity:** Position, tangent direction, and curvature continuous

25(c) at $t = 1$

Second derivatives:

Left: $(0, 6t) \rightarrow (0, 6)$

Right: $(12t^2, 2) \rightarrow (12, 2)$

✗ Second derivatives not equal: $(0, 6) \neq (12, 2)$

Curvature:

Different curvature on left and right sides

Answer: Not C(2) or G(2) continuous

25(d) at $t = 1$

Second derivatives:

Left: $(0, 2) \rightarrow (0, 2)$

Right: $(0, 2 + 6(t - 1)) \rightarrow (0, 2)$

✓ Second derivatives equal: $(0, 2) = (0, 2)$

Curvature:

Both sides have same position, first and second derivatives at junction

✓ Curvature continuous

Answer: C(2) continuous (and therefore G(2) continuous)

Function	C(2)	G(2)
25(c) at $t = 1$	No	No
25(d) at $t = 1$	Yes	Yes

Practice Problem 27: Lagrange Polynomial

- $P_0 = (0, 0)$
- $P_1 = (1, 2)$
- $P_2 = (2, 0)$

Lagrange Interpolation Formula:

$$P(x) = y_0 \cdot L_0(x) + y_1 \cdot L_1(x) + y_2 \cdot L_2(x)$$

$$L_k(x) = \prod_{\substack{i=0 \\ i \neq k}}^2 \frac{x - x_i}{x_k - x_i}$$

Step 1: Calculate Lagrange Basis Polynomials

For $L_0(x)$ (corresponding to $x_0 = 0$):

$$L_0(x) = \frac{(x - x_1)(x - x_2)}{(x_0 - x_1)(x_0 - x_2)} = \frac{(x - 1)(x - 2)}{(0 - 1)(0 - 2)} = \frac{(x - 1)(x - 2)}{2}$$

For $L_1(x)$ (corresponding to $x_1 = 1$):

$$L_1(x) = \frac{(x - x_0)(x - x_2)}{(x_1 - x_0)(x_1 - x_2)} = \frac{x(x - 2)}{(1 - 0)(1 - 2)} = \frac{x(x - 2)}{-1} = -x(x - 2)$$

For $L_2(x)$ (corresponding to $x_2 = 2$):

$$L_2(x) = \frac{(x - x_0)(x - x_1)}{(x_2 - x_0)(x_2 - x_1)} = \frac{x(x - 1)}{(2 - 0)(2 - 1)} = \frac{x(x - 1)}{2}$$

Step 2: Construct the Quadratic Polynomial

$$P(x) = y_0 \cdot L_0(x) + y_1 \cdot L_1(x) + y_2 \cdot L_2(x)$$

$$P(x) = 0 \cdot L_0(x) + 2 \cdot L_1(x) + 0 \cdot L_2(x)$$

$$P(x) = 2 \cdot [-x(x - 2)]$$

$$P(x) = -2x(x - 2)$$

$$P(x) = -2x^2 + 4x$$

Step 3: Verification

At x = 0:

$$P(0) = -2(0)^2 + 4(0) = 0 \quad \checkmark \text{ Matches } P_0 = (0, 0)$$

At x = 1:

$$P(1) = -2(1)^2 + 4(1) = -2 + 4 = 2 \quad \checkmark \text{ Matches } P_1 = (1, 2)$$

At x = 2:

$$P(2) = -2(2)^2 + 4(2) = -8 + 8 = 0 \quad \checkmark \text{ Matches } P_2 = (2, 0)$$

Final Answer:

$$\boxed{P(x) = -2x^2 + 4x}$$

This is the explicit quadratic representation of the curve passing through the given three points.